CSE 360 TP2 Tasks 6 & 7: Contributor Testing Framework & Test Javadocs

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1. Iterative Development & Testing Summary (Task 6 Feeding Into Task 7)

- **Iteration 1 — Entity & Collection Baseline Verification:** Inspected `entityClasses.LessonLearned` and `entityClasses.LessonLearnedList` in the TP2 repository. Confirmed that `LessonLearned` enforces title (4–64 chars), description (10–1000 chars), author, category, creation timestamp, and TP3 forward-compatibility attributes (`isFlagged`, `moderationNote`, `endorsementCount`, `isVerified`).
- **Iteration 2 — Pre-Mutation Input Validation & Ownership Guards:** Implemented `guiContributor.ContributorValidator` to validate boundary conditions and return helpful error messages before any collection or database mutation, and to enforce that only the original author or an Administrator can update or delete a lesson.
- **Iteration 3 — Semi-Automated & Manual Testbed Execution:** Built `testing.ContributorTestingAutomation` (28 automated positive and negative test cases; 28 passed, 0 failed) and `testing.ContributorManualTestbed` (5 interactive manual test procedures) modeled after the Foundations `PasswordEvaluationTestbed`. Generated official Javadoc HTML files and 33 source code screenshots in `ContributorTests_Javadocs/`.

2. TP2 Requirements Overview & Associated Javadoc / Screenshot Sets (Rubric 7.1 & 7.2)

Req ID & Story	TP2 Requirement Description	Assigned Tests	Javadoc HTML Files	Source Screenshot Set
TP2-REQ-01 US-HW2-01 (Create)	Create and insert a valid LessonLearned entity into LessonLearnedList while rejecting null references and duplicate primary key IDs.	TC-01, TC-02, TC-03, Manual-01	`testing/ContributorManualTestbed.html` `testing/ContributorTestingAutomation.html`	`TC01_CreateValidLesson_Screenshot.png` … (4 screenshots)
TP2-REQ-02 US-HW2-01 (Title Validation)	Validate that the LessonLearned title is non-null, non-blank, and between 4 and 64 characters after trimming whitespace.	TC-04, TC-05, TC-06	`testing/ContributorTestingAutomation.html`	`TC04_ValidateValidTitleBoundaries_Screenshot.png` … (3 screenshots)
TP2-REQ-03 US-HW2-01 (Description Validation)	Validate that the LessonLearned description body is non-null, non-blank, and between 10 and 1000 characters after trimming whitespace.	TC-07, TC-08, TC-09	`testing/ContributorTestingAutomation.html`	`TC07_ValidateValidDescriptionBoundaries_Screenshot.png` … (3 screenshots)

Req ID & Story	TP2 Requirement Description	Assigned Tests	Javadoc HTML Files	Source Screenshot Set
TP2-REQ-04 US-HW2-01 (Category & Author Validation)	Validate that the selected category is an approved technical domain (rejecting placeholders) and that the author username is 4 to 32 alphanumeric characters starting with a letter.	TC-10, TC-11	`testing/ContributorTestingAutomation.html`	`TC10_ValidateValidCategoryAndAuthor_Screenshot.png` … (2 screenshots)
TP2-REQ-05 US-HW2-02 (Read All & By ID)	Retrieve an unmodifiable view of all stored lessons via getAllLessons() and locate individual records by primary key ID via findById().	TC-12, TC-13, Manual-02	`testing/ContributorManualTestbed.html` `testing/ContributorTestingAutomation.html`	`TC12_ReadAllAndFindExistingId_Screenshot.png` … (3 screenshots)
TP2-REQ-06 US-HW2-02 (Keyword Search Subsets)	Execute case-insensitive keyword searches across title and description supporting arbitrary subset sizes (0, 1, or multiple elements) and rejecting queries over 64 characters.	TC-14, TC-15, TC-16	`testing/ContributorTestingAutomation.html`	`TC14_SearchKeywordMultiAndSingleSubset_Screenshot.png` … (3 screenshots)
TP2-REQ-07 US-HW2-02 (Category Filter Subsets)	Filter the master collection by exact category match into currentSubset, resetting to the full collection when 'All' or blank is provided.	TC-17, TC-18	`testing/ContributorTestingAutomation.html`	`TC17_FilterCategoryMatchingAndAllReset_Screenshot.png` … (2 screenshots)
TP2-REQ-08 US-HW2-03 (Update Lesson & Sync)	Update mutable fields (title, description, category) of an existing lesson by primary key ID, enforcing boundary validation and synchronizing active subsets.	TC-19, TC-20, Manual-03	`testing/ContributorManualTestbed.html` `testing/ContributorTestingAutomation.html`	`TC19_UpdateValidLessonAndSyncSubset_Screenshot.png` … (3 screenshots)
TP2-REQ-09 US-HW2-03 (Update Authorization)	Enforce ownership authorization so only the original contributor author or an Administrator can modify an existing lesson learned.	TC-21, TC-22	`testing/ContributorTestingAutomation.html`	`TC21_AuthorizeUpdateByOriginalAuthorAndAdmin_Screenshot.png` … (2 screenshots)
TP2-REQ-10 US-HW2-04 (Delete Lesson & Sync)	Permanently expel a lesson learned by primary key ID from both the master collection and active subset, rejecting non-existent or non-positive IDs.	TC-23, TC-24, Manual-04	`testing/ContributorManualTestbed.html` `testing/ContributorTestingAutomation.html`	`TC23_DeleteValidLessonFromMasterAndSubset_Screenshot.png` … (3 screenshots)

Req ID & Story	TP2 Requirement Description	Assigned Tests	Javadoc HTML Files	Source Screenshot Set
TP2-REQ-11 US-HW2-04 (Delete Authorization)	Enforce ownership authorization so only the original contributor author or an Administrator can delete an existing lesson learned.	TC-25, TC-26	`testing/ContributorTestingAutomation.html`	`TC25_AuthorizeDeleteByOriginalAuthorAndAdmin_Screenshot.png` … (2 screenshots)
TP2-REQ-12 TP3 Epics Forward Compatibility (Rubric 3.3)	Support forward-compatible TP3 governance operations: administrative moderation flagging (setModerationFlag), peer endorsement increments (incrementEndorsement), and staff verification (setTechnicalVerification).	TC-27, TC-28, Manual-05	`testing/ContributorManualTestbed.html` `testing/ContributorTestingAutomation.html`	`TC27_VerifyTP3ModerationEndorsementAndVerification_Screenshot.png` … (3 screenshots)

3. Complete Test-by-Test Javadoc, Screenshot, and Output Interpretation Matrix (Rubric 7.2, 7.3 & 7.4)

Req & Test ID	Test Method Name	Javadoc HTML File	Source Screenshot File	How Code Satisfies Requirement & Output Interpretation
TP2-REQ-01 TC-01 (Positive)	`performTestCase01_CreateValidLesson()`	`testing/ContributorTestingAutomation.html`	`TC01_CreateValidLesson_Screenshot.png`	Instantiates a valid LessonLearned (#101) and calls addLesson(); passes when addLesson() returns true and getTotalCount() == 1.
TP2-REQ-01 TC-02 (Negative)	`performTestCase02_CreateNullLesson()`	`testing/ContributorTestingAutomation.html`	`TC02_CreateNullLesson_Screenshot.png`	Passes null to addLesson(null); passes when defensive guard returns false and getTotalCount() == 0.
TP2-REQ-01 TC-03 (Negative)	`performTestCase03_CreateDuplicateIdLesson()`	`testing/ContributorTestingAutomation.html`	`TC03_CreateDuplicateIdLesson_Screenshot.png`	Attempts to add a second LessonLearned with duplicate ID #101; passes when second addLesson() returns false and count remains 1.
TP2-REQ-01 Manual-01 (Pos/Neg)	`executeManualCreateTest()`	`testing/ContributorManualTestbed.html`	`Manual01_ExecuteManualCreateTest_Screenshot.png`	Interactive manual test validating fields via validateLessonFields() before calling addLesson().

Req & Test ID	Test Method Name	Javadoc HTML File	Source Screenshot File	How Code Satisfies Requirement & Output Interpretation
TP2-REQ-02 TC-04 (Positive)	`performTestCase04_ValidateValidTitleBoundaries()`	`testing/ContributorTestingAutomation.html`	`TC04_ValidateValidTitleBoundaries_Screenshot.png`	Tests 4-char lower bound, standard length, and 64-char upper bound in ContributorValidator.validateTitle(); passes when all return empty string ''.
TP2-REQ-02 TC-05 (Negative)	`performTestCase05_ValidateBlankAndShortTitle()`	`testing/ContributorTestingAutomation.html`	`TC05_ValidateBlankAndShortTitle_Screenshot.png`	Tests whitespace title (' ') and 3-char title ('Bug'); passes when specific 'cannot be blank' and 'too short' error strings are returned.
TP2-REQ-02 TC-06 (Negative)	`performTestCase06_ValidateOversizedTitle()`	`testing/ContributorTestingAutomation.html`	`TC06_ValidateOversizedTitle_Screenshot.png`	Tests 65-character title; passes when validateTitle() returns error stating title is too long (must be 4 to 64 chars).
TP2-REQ-03 TC-07 (Positive)	`performTestCase07_ValidateValidDescriptionBoundaries()`	`testing/ContributorTestingAutomation.html`	`TC07_ValidateValidDescriptionBoundaries_Screenshot.png`	Tests 10-char minimum and 1000-char maximum descriptions in ContributorValidator.validateDescription(); passes when both return ''.
TP2-REQ-03 TC-08 (Negative)	`performTestCase08_ValidateBlankAndShortDescription()`	`testing/ContributorTestingAutomation.html`	`TC08_ValidateBlankAndShortDescription_Screenshot.png`	Tests empty string ('') and 9-char string ('ShortDesc'); passes when specific 'cannot be empty' and 'too short' errors are returned.
TP2-REQ-03 TC-09 (Negative)	`performTestCase09_ValidateOversizedDescription()`	`testing/ContributorTestingAutomation.html`	`TC09_ValidateOversizedDescription_Screenshot.png`	Tests 1001-character description; passes when validateDescription() returns error stating description is too long.
TP2-REQ-04 TC-10 (Positive)	`performTestCase10_ValidateValidCategoryAndAuthor()`	`testing/ContributorTestingAutomation.html`	`TC10_ValidateValidCategoryAndAuthor_Screenshot.png`	Tests approved category ('Architecture'), valid author ('Sanidhya26'), and validateLessonFields(); passes when all return ''.

Req & Test ID	Test Method Name	Javadoc HTML File	Source Screenshot File	How Code Satisfies Requirement & Output Interpretation
TP2-REQ-04 TC-11 (Negative)	`performTestCase11_ValidateInvalidCategoryAndAuthor()`	`testing/ContributorTestingAutomation.html`	`TC11_ValidateInvalidCategoryAndAuthor_Screenshot.png`	Tests placeholder ('Select Category'), unapproved category ('Astrology'), digit-leading author ('9User'), and blank author (''); passes when all return error messages.
TP2-REQ-05 TC-12 (Positive)	`performTestCase12_ReadAllAndFindExistingId()`	`testing/ContributorTestingAutomation.html`	`TC12_ReadAllAndFindExistingId_Screenshot.png`	Reads 3 stored lessons via getAllLessons() and queries ID #202 via findById(202); passes when size == 3 and Optional is present.
TP2-REQ-05 TC-13 (Negative)	`performTestCase13_ReadNonExistentIdAndUnmodifiableView()`	`testing/ContributorTestingAutomation.html`	`TC13_ReadNonExistentIdAndUnmodifiableView_Screenshot.png`	Queries missing ID #9999 and attempts getAllLessons().clear(); passes when Optional.empty() is returned and UnsupportedOperationException blocks direct mutation.
TP2-REQ-05 Manual-02 (Pos/Neg)	`executeManualReadAndSearchTest()`	`testing/ContributorManualTestbed.html`	`Manual02_ExecuteManualReadAndSearchTest_Screenshot.png`	Interactive manual test reading and searching lessons in currentSubset.
TP2-REQ-06 TC-14 (Positive)	`performTestCase14_SearchKeywordMultiAndSingleSubset()`	`testing/ContributorTestingAutomation.html`	`TC14_SearchKeywordMultiAndSingleSubset_Screenshot.png`	Searches 'javafx' (multi-item subset = 2) and 'branch' (single-item subset = 1); passes when subset counts match and isSubsetEmpty() is false.
TP2-REQ-06 TC-15 (Negative)	`performTestCase15_SearchKeywordEmptySubset()`	`testing/ContributorTestingAutomation.html`	`TC15_SearchKeywordEmptySubset_Screenshot.png`	Searches unmatched keyword 'nonexistent_quantum_kernel'; passes when getSubsetSize() == 0 and isSubsetEmpty() == true without null exceptions.

Req & Test ID	Test Method Name	Javadoc HTML File	Source Screenshot File	How Code Satisfies Requirement & Output Interpretation
TP2-REQ-06 TC-16 (Negative)	`performTestCase16_SearchKeywordBlankResetAndOversizedQuery()`	`testing/ContributorTestingAutomation.html`	`TC16_SearchKeywordBlankResetAndOversizedQuery_Screenshot.png`	Passes 65-char query to validateSearchKeyword() and blank query to searchByKeyword(); passes when 65-char query is rejected and blank query resets subset.
TP2-REQ-07 TC-17 (Positive)	`performTestCase17_FilterCategoryMatchingAndAllReset()`	`testing/ContributorTestingAutomation.html`	`TC17_FilterCategoryMatchingAndAllReset_Screenshot.png`	Calls filterByCategory('GUI') (1 match) and filterByCategory('All') (3 matches); passes when subset counts equal 1 and 3.
TP2-REQ-07 TC-18 (Negative)	`performTestCase18_FilterCategoryNoMatchEmptySubset()`	`testing/ContributorTestingAutomation.html`	`TC18_FilterCategoryNoMatchEmptySubset_Screenshot.png`	Calls filterByCategory('Testing') when no records match; passes when getSubsetSize() == 0 and isSubsetEmpty() == true.
TP2-REQ-08 TC-19 (Positive)	`performTestCase19_UpdateValidLessonAndSyncSubset()`	`testing/ContributorTestingAutomation.html`	`TC19_UpdateValidLessonAndSyncSubset_Screenshot.png`	Updates Lesson #201 via updateLesson() while present in currentSubset; passes when true is returned and both master and subset reflect the new title.
TP2-REQ-08 TC-20 (Negative)	`performTestCase20_UpdateInvalidFieldsAndNonExistentId()`	`testing/ContributorTestingAutomation.html`	`TC20_UpdateInvalidFieldsAndNonExistentId_Screenshot.png`	Attempts updateLesson() with 2-char title, 5-char description, blank category, and non-existent ID #9999; passes when all return false and record #201 is unchanged.
TP2-REQ-08 Manual-03 (Pos/Neg)	`executeManualUpdateTest()`	`testing/ContributorManualTestbed.html`	`Manual03_ExecuteManualUpdateTest_Screenshot.png`	Interactive manual test checking ID existence, ownership permission, and field validation before calling updateLesson().
TP2-REQ-09 TC-21 (Positive)	`performTestCase21_AuthorizeUpdateByOriginalAuthorAndAdmin()`	`testing/ContributorTestingAutomation.html`	`TC21_AuthorizeUpdateByOriginalAuthorAndAdmin_Screenshot.png`	Checks verifyModifyPermission() on Lesson #201 for author 'Sanidhya' and Admin 'AdminPranjal'; passes when both return ''.

Req & Test ID	Test Method Name	Javadoc HTML File	Source Screenshot File	How Code Satisfies Requirement & Output Interpretation
TP2-REQ-09 TC-22 (Negative)	`performTestCase22_RejectUpdateByUnauthorizedContributor()`	`testing/ContributorTestingAutomation.html`	`TC22_RejectUpdateByUnauthorizedContributor_Screenshot.png`	Checks verifyModifyPermission() on 'Sanidhya' lesson for non-admin contributor 'PeerUserBob'; passes when 'Unauthorized:' error is returned.
TP2-REQ-10 TC-23 (Positive)	`performTestCase23_DeleteValidLessonFromMasterAndSubset()`	`testing/ContributorTestingAutomation.html`	`TC23_DeleteValidLessonFromMasterAndSubset_Screenshot.png`	Deletes Lesson #202 via deleteLesson(202) while in currentSubset; passes when true is returned, totalCount drops to 2, and subsetSize drops to 0.
TP2-REQ-10 TC-24 (Negative)	`performTestCase24_DeleteNonExistentAndNegativeId()`	`testing/ContributorTestingAutomation.html`	`TC24_DeleteNonExistentAndNegativeId_Screenshot.png`	Calls deleteLesson() with IDs 9999, 0, and -5; passes when all return false and totalCount remains 3.
TP2-REQ-10 Manual-04 (Pos/Neg)	`executeManualDeleteTest()`	`testing/ContributorManualTestbed.html`	`Manual04_ExecuteManualDeleteTest_Screenshot.png`	Interactive manual test checking ID existence and ownership permission before calling deleteLesson().
TP2-REQ-11 TC-25 (Positive)	`performTestCase25_AuthorizeDeleteByOriginalAuthorAndAdmin()`	`testing/ContributorTestingAutomation.html`	`TC25_AuthorizeDeleteByOriginalAuthorAndAdmin_Screenshot.png`	Verifies delete permission on Lesson #202 for author 'Gauri' and Admin 'AdminUser' and executes deleteLesson(202); passes when authorized and deleted.
TP2-REQ-11 TC-26 (Negative)	`performTestCase26_RejectDeleteByUnauthorizedContributor()`	`testing/ContributorTestingAutomation.html`	`TC26_RejectDeleteByUnauthorizedContributor_Screenshot.png`	Attempts delete check on 'Gauri' lesson as non-owner 'Sanidhya' and as null user; passes when 'Unauthorized:' error is returned and record remains.
TP2-REQ-12 TC-27 (Positive)	`performTestCase27_VerifyTP3ModerationEndorsementAndVerification()`	`testing/ContributorTestingAutomation.html`	`TC27_VerifyTP3ModerationEndorsementAndVerification_Screenshot.png`	Invokes setModerationFlag, incrementEndorsement (twice), and setTechnicalVerification on Lesson #201; passes when flag=true, note trimmed, count=2, and verified=true.

Req & Test ID	Test Method Name	Javadoc HTML File	Source Screenshot File	How Code Satisfies Requirement & Output Interpretation
TP2-REQ-12 TC-28 (Negative)	`performTestCase28_RejectTP3HooksOnNonExistentId()`	`testing/ContributorTestingAutomation.html`	`TC28_RejectTP3HooksOnNonExistentId_Screenshot.png`	Invokes all three TP3 hooks on non-existent ID #9999; passes when all three return false.
TP2-REQ-12 Manual-05 (Pos/Neg)	`executeManualTP3GovernanceTest()`	`testing/ContributorManualTestbed.html`	`Manual05_ExecuteManualTP3GovernanceTest_Screenshot.png`	Interactive manual test exercising TP3 moderation, endorsement, and staff verification hooks.